

NLP - Mid sem 2025-26 Closed book - Regular

Question 1

You can write your answers in the provided space or write your answer on a piece of paper and scan and upload the handwritten answers using the QR Code available in the Scan and Upload Section of this exam.

Kindly ensure all answer sheets to be uploaded against the corresponding questions only. All sheets should not be uploaded against one question.

A) Identify the suitable NLP tasks for the use cases what we use in our daily life [3 Marks]

- a. Auto suggestion of next tokens while typing email or whatsapp
- b. Identifying the company names, drug names, quarterly revenue from the business article of a pharma company
- c. Converting the message posted on Facebook in Hindi language to English language

B) What is linearly inseparability and illustrate it with the implementation of XOR using perceptions [2 Marks]

Question 2

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A) Consider the following statements: [2 Marks]

- a) *The robot moved towards the right*
- b) *The robot turned left*
- c) *The robot stopped suddenly*

Apply basic pre-processing techniques such as lower case. (i.e. DO NOT remove stop words / lemmatization / stemming / etc.) carry out the following tasks:

- i) Create a vocabulary from the sentences
- ii) Compute the TF-IDF representation for each word

B) Consider the following text corpus: [2 Marks]

<s>Science is interesting</s>

<s>History is easy</s>

<s>Physics is part of Science</s>

Compute bigram probabilities of $P(\text{is}|\text{Science})$ and $P(\text{easy}|\text{is})$

Question 3

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A) There is a document collection of 1000 documents including 500 documents of biology and 500 documents of AI. The word 'blood' occurs in 100 documents (of biology) and word 'success' occurs in 5 documents of biology and 5 documents of AI. [3 Marks]

- a. Compare the IDF of both words.
- b. Comment on results.

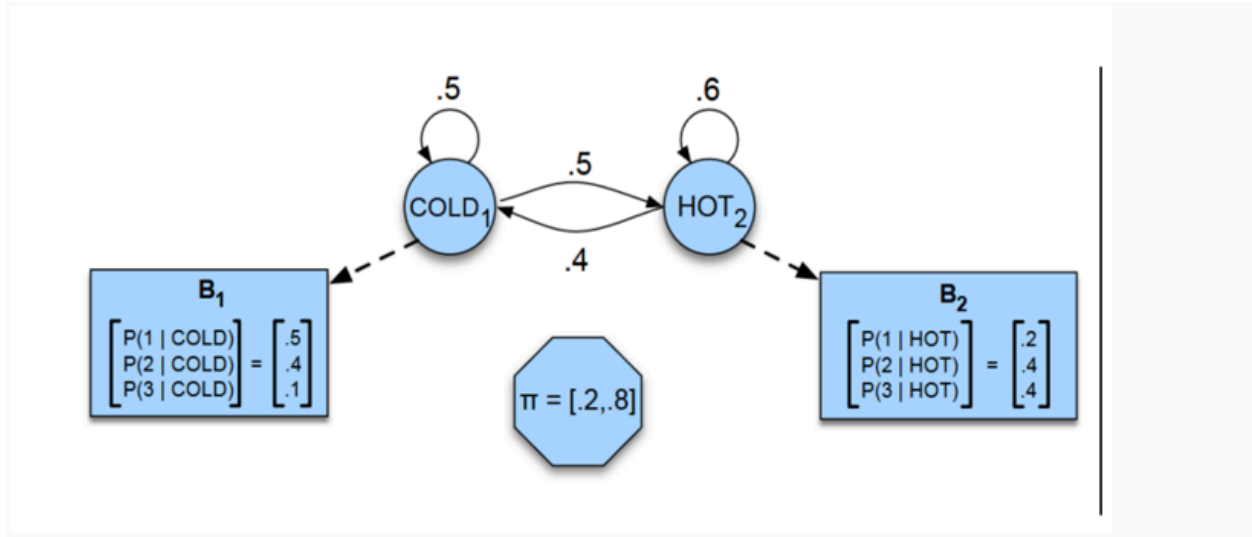
B) In a large corpus, words and their frequencies are computed. The word 'in' occurs 1.25% in the corpus and it is the 6th most frequent word. The word 'that' is the 10th most frequent word. As per Zipf's law, what is the expected frequency of the word? [2 Marks]

Question 4

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Following HMM diagram shows the probabilities of the number of ice creams eaten by X on hot/cold days. Compute the hidden sequence(of weather) generated for the sequence of (1, 3, 3) ice creams eaten by 'X' (the observations) on consecutive days. [5 Marks]



Question 5

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I)

- Illustrate why NLP is hard?
- Differentiate lemmatization and stemming.
- How can morphology help with NLP tasks?

[3 Marks]

II) A restaurant serves 10 dishes. They have 5 servers, and all customers know their names.

The conversations refer to each dish name 5% times, and each server name 10% times (total 100%, ignoring other words). Compute perplexity of conversations in the restaurant. [3 Marks]